MindWell: A Conversational Agent for Professional Depression Screening on Social Media

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Abstract. Depression is among the most prevalent mental health conditions, with an early and accurate diagnosis being essential for mitigating its effects. Yet, stigma often prevents individuals from seeking professional help. In this context, social media offers a unique resource for depression screening, as users frequently share, comment, and disclose their daily struggles, providing key insights into their mental health through online activity. However, the immense volume of data generated on these platforms presents a significant challenge, requiring substantial time and effort for mental health professionals to analyze. This demo paper introduces MindWell, an open-source conversational agent designed to support clinicians in identifying symptoms and emotions relevant to clinical assessments. MindWell uses a Retrieval-Augmented Generation (RAG) framework, incorporating a Large Language Model (LLM) based on Llama 3.1 and fine-tuned specifically for depression screening based on clinical symptom criteria, particularly the Beck Depression Inventory-II (BDI-II). By leveraging users' social media history as informed and reliable context, MindWell is designed to answer questions formulated by clinicians, facilitating the review process. We collaborated with a professional psychologist to assess MindWell's responses in a clinical setting, finding that the system effectively captures users' depressive signs and shows promise for mental health support applications.

Keywords: Depression Detection · Conversational AI · Explainable AI · Large Language Model · Retrieval Augmented Generation · Open Source

1 Background and Motivation

Mental health is essential for overall well-being, yet over 20% of adults experience mental disorders [14,2]. Depression alone affects approximately 332 million people worldwide [11]. Unfortunately, many individuals at risk do not seek help due to stigma, with over 60% of those experiencing depression avoiding professional care because of societal perceptions [9,16]. Additional barriers, such as financial constraints, lack of insurance, and limited access to mental health resources, especially in lower socioeconomic areas, further restrict access to mental

health care [6]. In this context, social media emerges as a promising resource for detecting signs of mental disorders, as individuals often share their thoughts and emotions openly online. This online activity can provide valuable insights into users' mental states [5,7]. However, the immense data generated on social platforms presents a significant challenge, requiring extensive time and resources for mental health professionals to analyze effectively.

To address this challenge, we propose MindWell, a conversational agent designed to assist mental health professionals in the task of depression screening. Previous studies have shown that analyzing social media content can help in identifying mental health conditions such as depression [15,12,19]. In line with this research, and using Reddit as the target social platform, we have developed a Retrieval-Augmented Generation (RAG) [10] system that processes users social media history as informed context. Using our application, mental health professionals can ask MindWell about specific symptoms or emotions relevant to clinical diagnosis, facilitating the time-consuming process of reviewing social media data manually. This allows clinicians to screen larger populations more efficiently and to focus early on those users at higher risk.

To better align the system with mental health professional needs [3,18], we fine-tuned Llama 3.1 to provide contextually relevant responses that identify depression indicators according to the Beck Depression Inventory-II (BDI-II)[4]. As a result, the LLM that is part of our system can be better interpretable by clinicians. This work builds on the research by Pérez et al.[13], who developed a platform for examining and tracking depression symptoms on social media, supporting professionals through data labeling and depression profiling tools. We extend this foundation by integrating automated social media analysis with conversational capabilities, enabling MindWell to detect depressive signs in user posts. Our source code is open-source at https://gitlab.irlab.org/ecir-mindwell, and a demonstration video is available at https://www.irlab.org/mindwell-demo.mp4.

Our work falls within the recent explosion of chatbot applications, driven by the popularization of conversational LLMs. Chatbots have been deployed across a wide range of tasks, including mental health support [8]. However, existing conversational agents, such as *Woebot*[1] and *Wysa*[2], were designed primarily for direct user interaction and do not provide tools for professionals to analyze social media content. In contrast, our approach supports depression screening by facilitating a conversational agent to assist in the analysis of social media activity, alleviating the complex task of manually reviewing posts to identify evidence or signs of depressive symptoms.

2 System Overview

MindWell is an iteration based on the prior work of Pérez et al. [13]. Thereby, the core architecture is inherited from the already existing platform. We also

[1] https://woebothealth.com
[2] https://www.wysa.com

made use of existing module responsible for obtaining and storing all the writing history from Reddit users for data analysis. Figure 1 provides an overview of the system architecture, where new components highlighted in green and modifications to existing elements shown in blue.

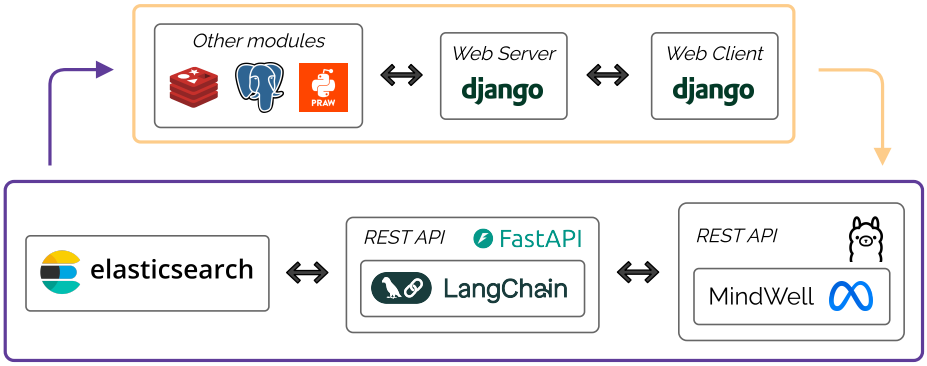


Fig. 1. System architecture overview. New components introduced in MindWell are highlighted in purple, while modifications to existing elements from PsyProf are shown in orange.

Modifications to existing modules consisted of extending *Django* views and controllers to include the necessary components for the chat interface. Additionally, we modified the persistence and caching modules to store conversation history using *PostgreSQL*. As for the new components created specifically for the RAG pipeline, we divide them into the following blocks: **(1) LLM.** Our tool is based on a fine-tuned `Llama-3.1-8B-Instruct`. The model is deployed via an Application Programming Interface (API) that locally replicates the OpenAI API. This allows the model to be consumed on HTTP requests. **(2) *Elasticsearch*[3] index.** Users' social media writings are organized within an Elasticsearch index. Specifically, we compute vector embeddings for these comments using the `sentence-transformers/all-mpnet-base-v2` model and store them along with metadata including ID, publication date, subreddit, and a link to the writing. Therefore, when a clinician poses a question to the chatbot, the system uses this question as a query to perform a semantic similarity search within the index. Finally, the top 25 most relevant posts to the query are retrieved as context for generating responses. **(3) *LangChain*[4].** This module serves as an intermediary between the LLM and the *Elasticsearch* index. We implemented it as an API that receives requests from the *Django* view. It then uses *LangChain* chains to communicate with the index to retrieve the context for inference, build the query,

[3] https://www.elastic.co/es/elasticsearch
[4] https://www.langchain.com module

and finally call the model API to ultimately get the model response based on that context.

3 Evaluation

To evaluate the effectiveness of the conversational agent as a support tool for mental health professionals, we collaborated with a clinical psychologist. After conducting training sessions to familiarize the practitioner with the application, we collected detailed feedback on three key aspects:

(1) Usability: The psychologist rated the application's usability as high, describing that the interface was intuitive and easy to use. However, they suggested a brief onboarding tutorial for new users to help them get started more efficiently. **(2) Utility:** The psychologist found the application to be highly useful in a depression screening scenario. They noted that MindWell effectively identified patterns of depressive symptoms by analyzing the user's social media posts, providing insights that aligned with standard clinical assessments. The practitioner highlighted that the system's ability to detect shifts in language and tone over time could be particularly valuable for early diagnosis and monitoring of depressive symptoms. **(3) Suggestions for Improvement:** While the overall feedback was positive, the psychologist recommended a few enhancements. They suggested improving the explanation feature, allowing clinicians to see a more detailed breakdown of how the system derived its conclusions, which would strengthen the trust and interpretability of the results. Additionally, they proposed integrating more specific markers or questionnaires beyond the BDI-II.

4 Conclusions and Ethical Statement

Through this work, we are open-sourcing MindWell, a conversational agent that has shown promise as a support tool for mental health professionals. By analyzing users' social media posts, MindWell offers insights aligned with clinically validated markers of depressive symptoms, answering queries about patterns and emotional states. This functionality not only facilitates the labor-intensive task of reviewing social media histories but also helps clinicians identify user behaviors that might otherwise go unnoticed. Positive feedback from the psychologist involved suggests that MindWell effectively complements clinical assessments, enabling more timely and informed interventions.

For future research, we plan to extend MindWell to other social networks, addressing the challenge of cross-platform generalization. Additionally, based on professional feedback, we aim to incorporate further psychological assessments beyond the BDI-II, such as the DSM-5[1]. As a concluding note, we emphasize that MindWell is intended to complement rather than replace mental health professionals and requires validation and oversight by licensed clinicians[17]. We collaborate closely with healthcare professionals at every stage and will continue to do so to ensure its clinical alignment.

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